NBC, NUST

BS Artificial Intelligence · 2024F · Section A · Fall 2026

DS-311 DATA MINING

Semester Project Proposal

Student Success Early-Warning

Project P10 · Milestone M0

A data-mining study that identifies students at risk of dropping out early enough to help, and uncovers the patterns that come before dropout.

SUPERVISION

Course Instructor	Dr. Zirak Khan
Lab Instructor	Syed Daniyal Asghar
Course	DS-311 Data Mining · Semester Project (Week 6 Lab)

GROUP MEMBERS

Group Number	03

#	Student Name	Registration No.	Project Role
1	Areeba Zahra	514014	Mining Lead: modelling and evaluation; implements classification
2	Qudsia Batool	501696	Data Lead: acquisition, data-quality report, EDA; implements clustering
3	Madeeha Noor	529378	Product Lead: web app, dashboard, report; implements association rules

PROJECT LINKS

GitHub Repository	https://github.com/madeeha-noor/Student-Success-Early-Warning
Colab Notebook	notebooks/01_data_loading.ipynb (open in Colab)

Submission date: 14 October 2026 (Week 6 Lab)

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Overview

Project	P10 Student Success Early-Warning (Group 03)
Question	Can we identify students at risk of dropping out early enough to help, and which patterns come before dropout?
Dataset	UCI “Predict Students’ Dropout and Academic Success” (ID 697): 4,424 students, 36 features, three classes
Three questions	Q1 How early can we tell? · Q2 Which patterns come before dropout? · Q3 Who are the students at risk?
Deliverables	Colab notebook, GitHub repository, Streamlit advisor dashboard, 5–6 page report, presentation, live demo, individual viva
Repository	https://github.com/madeeha-noor/Student-Success-Early-Warning
Submission	14 October 2026 (Week 6 Lab, Milestone M0)

PROJECT PROPOSAL · P10 Student Success Early-Warning

DS-311 Data Mining · BS-AI 2024F, Section A · NBC, NUST · Instructor: Dr. Zirak Khan · Lab: Syed Daniyal Asghar · Group 03.

1. PROBLEM STATEMENT AND MOTIVATION

University dropout harms students, families and institutions, and it is most preventable when risk is spotted early. Yet outcomes are usually known only after a student has left. This project asks: **can we identify students at risk of dropping out early enough to help, and which patterns come before dropout?** We will build an end-to-end data-mining solution that flags at-risk students, explains the reasons behind each flag, and presents the results in an advisor dashboard, so that support (financial counselling, tutoring) can be offered while it can still make a difference.

2. DATASET

UCI "Predict Students' Dropout and Academic Success" (ID 697, Realinho et al., 2021; **CC BY 4.0**): **4,424 students, 36 features** from a higher-education institution: demographics, socio-economic and macro-economic indicators, admission details, and first- and second-semester curricular results. The target has three classes: **Graduate (2,209), Dropout (1,421), Enrolled (794)**, a moderate imbalance that we handle with stratified splits, class weights and macro-F1. The data is anonymised, loads directly in Colab through `ucimlrepo` (fetch_ucirepo(id=697)), and is small enough for free Colab. We will cite the source and licence. **Optional expert extension:** OULAD clickstream data for weekly-engagement prediction, attempted only after the core and intermediate work is complete.

3. KEY QUESTIONS AND HOW WE WILL ANSWER THEM

Research question	How the project will address it	How the answer is validated
Q1. How early can we tell? Can dropout be predicted from admission-time information alone, and how much does first-semester performance improve it?	Train the same models on two feature sets: (A) admission/demographic/socio-economic only, (B) A + first-semester results. Compare decision tree and Naive Bayes baselines with random forest and SVM. Feature selection (mutual information, RF importance) shows which few variables matter.	Stratified 5-fold CV inside scikit-learn Pipelines; macro-F1, per-class recall; majority-class baseline; leakage check; confusion matrix on the 'Enrolled' class.
Q2. Which patterns come before dropout? Which combinations of financial, academic and background factors co-occur with dropout?	Discretise key variables, then mine association rules with Apriori and FP-Growth (mlxtend) targeting the Dropout class, e.g. debtor ∧ tuition fees not up to date → Dropout. Chi-square tests confirm key relationships.	Support, confidence and lift with justified thresholds (not library defaults); Apriori vs. FP-Growth runtime and rule comparison; rules reported only if lift > 1 and interpretable.
Q3. Who are the students at risk? What distinct student profiles exist, and which carry the highest dropout rate?	Standardise and encode features, reduce with PCA, then cluster with K-means and hierarchical clustering. Name each profile and compute its Dropout/Enrolled/Graduate mix to rank profiles by risk.	Elbow and silhouette for k; K-means vs. hierarchical comparison; cluster stability across seeds; profile cards with dropout rate per cluster.

4. METHODOLOGY AND DELIVERABLES

Pipeline: acquire and decode categorical codes → data-quality report (missing values, duplicates, inconsistent values, outliers, each decision justified) → EDA with at least one hypothesis test (e.g. dropout vs. tuition-fee status) → encoding, scaling, discretisation, PCA / feature selection → mining (classification, association rules, clustering, each against a simple baseline) → fair evaluation with fixed seeds → interpretation and error analysis.

Deliverables: Colab notebook, GitHub repository (commits from all three members, README, requirements.txt), a **Streamlit advisor dashboard** (at-risk list with reasons, what-if sliders, cohort profile cards), 5–6 page report, presentation, live demo and individual viva.

Project links: GitHub repository https://github.com/madeeha-noor/Student-Success-Early-Warning · Colab notebook notebooks/01_data_loading.ipynb

5. TEAM ROLES AND TIMELINE

Mining Lead (Member 1): pipelines, baselines, model comparison, evaluation; implements classification (Q1).	**Data Lead (Member 2):** data loading, quality report, EDA, hypothesis tests; implements clustering (Q3).	**Product Lead (Member 3):** Streamlit app, dashboard, deployment, report; implements association rules (Q2).

Milestones: M0 Proposal (Wk 6, 14 Oct) · M1 Data quality & EDA (Wk 7, 21 Oct) · M2 Transformations and first baseline (Wk 8, 28 Oct) · M3 Core complete and app skeleton (Wk 11, 18 Nov) · M4 Full evaluation and deployed app (Wk 13, 2 Dec) · Feature freeze (Wk 15, 16 Dec) · Submission, demo and viva (Wk 16–17).

6. ETHICS, RISKS AND EXPECTED OUTCOME

Labelling students is sensitive. The model will be positioned as **decision support for advisors, never an automatic decision,** and we will audit error rates across gender, age group and scholarship status and discuss fairness and appropriate use in the report. Other risks are the 'Enrolled' class being hard to separate and leakage from second-semester features, both addressed by explicit checks.

Expected outcome: a validated model showing how early risk can be detected, a ranked set of interpretable dropout rules, named student risk profiles, and a working public app. Any generative-AI assistance will be disclosed per course policy.

Reference: Realinho, V., Machado, J., Baptista, L., & Martins, M. V. (2021). Predict Students' Dropout and Academic Success. UCI Machine Learning Repository. https://doi.org/10.24432/C5MC89 · Licence: CC BY 4.0